Q. 1 (Twitter)**.** For this problem, we need to convert rates to rates to proper units. On the hour timescale, $\mu_T = 4$, $\mu_J = 12$, and $\mu_O = 1$. On the minute timescale, we have $\mu_T = 1/15$, $\mu_J = 1/5$, and $\mu_O = 1/60$. Either of these would be accepted, but this solution will be written on the hour timescale.

a. For this problem, we superposition the Trump process and the Bieber process. By the principle of superposition, the probability that a given arrival is a Trump tweet is $\frac{\mu_T}{\mu_T+\mu_B} = \frac{3}{4}$, and similarly, it is a Bieber tweet with probability $\frac{\mu_B}{\mu_T+\mu_B} = \frac{1}{4}$.

   We want there to be at least one Trump tweet in the first five tweets. What happens afterwards is of no consequence. The most straightforward approach is to compute the probability there are no Trump tweets in the first five tweets, and subtract from 1. This gives directly $1 - \left(\frac{3}{4}\right)^5$.

   Alternatively, you could sum over when the Trump tweet arrives. For the conditions of our event to be satisfied, the first Trump tweet must happen in the first five tweets.

$$\begin{aligned}
\mathbf{P}(\text{Trump tweet in first five tweets}) &= \sum_{i=1}^{5} \mathbf{P}(\text{First Trump tweet at time } i) \\
&= \sum_{i=1}^{5} \left(\frac{3}{4}\right)^{i-1} \cdot \frac{1}{4} \\
&= \frac{1}{4} \cdot \frac{1 - \left(\frac{3}{4}\right)^5}{1 - \frac{3}{4}} \\
&= 1 - \left(\frac{3}{4}\right)^5
\end{aligned}$$

   It is also possible (but not recommended) to integrate over all values of t at which the fifth Bieber tweet could come. This random variable, $T_5^{(J)}$ follows a $\Gamma(5, 12)$ distribution, and find the probability that the first Trump tweet time ($T_1^{(T)} \sim Exp(4)$) comes before that time. We set up the integral here, and simplify by manipulating the integrand to be gamma distribution.

$$\begin{aligned}
\mathbf{P}(T_1^{(T)} < T_5^{(J)}) &= \int_0^\infty \mathbf{P}(T_1^{(T)} < T_5^{(J)} \mid T_5^{(J)} = t) \cdot \mathbf{P}(T_5^{(J)} \in dt) \\
&= \int_0^\infty \mathbf{P}(T_1^{(T)} < t) \cdot \mathbf{P}(T_5^{(J)} \in dt) \\
&= \int_0^\infty (1 - e^{-4t}) \cdot \frac{12^5 t^4 e^{-12t}}{4!} dt \\
&= \int_0^\infty \frac{12^5 t^4 e^{-12t}}{4!} dt - \int_0^\infty \frac{12^5 t^4 e^{-16t}}{4!} dt \\
&= 1 - \left(\frac{12}{16}\right)^5 \int_0^\infty \frac{16^5 t^4 e^{-16t}}{4!} dt \\
&= 1 - \left(\frac{3}{4}\right)^5
\end{aligned}$$

b. We seek the expected time of the first tweet given that Trump tweeted first. Let $T_1 \sim \text{Exp}(17)$ be the time of the first tweet, and X_1 be the label of the first tweet. By the principle of superposition, the time of the first tweet T_1 (which is a function of the superposed process) is independent of "Trump tweets first" label, so

$$\mathbf{E}[T_1 \mid X_1 = (T)] = E[T_1] = \frac{1}{17}.$$

c. We want the expected time that you see a Bieber tweet *and* an Oprah tweet. Let $T_1^{(B)}, T_1^{(O)}$ be the first tweet times of Bieber and Oprah respectively. We seek $\mathbf{E}[T_1^{(B)} \vee T_1^{(O)}]$. One approach, using the fact that $X \vee Y + X \wedge Y = X + Y$ is

$$\mathbf{E}[T_1^{(B)} \vee T_1^{(O)}] = \mathbf{E}[T_1^{(B)}] + \mathbf{E}[T_1^{(O)}] - \mathbf{E}[T_1^{(B)} \wedge T_1^{(O)}] = 1 + \frac{1}{12} - \frac{1}{13}$$

where we have used the fact that $T_1^{(B)} \wedge T_1^{(O)} \sim \text{Exp}(13)$

Another approach is to find the expected time of first arrival in the Bieber, Oprah process, and then condition on the label of the first arrival and find the first time of arrival of the other label in the restarted process. The rate of the superpositioned process is $\mu_{BO} = 13$, so the expected first time of arrival is $\frac{1}{13}$. Next we condition on the first label and find the expected time of arrival of the other label. We get

$$\begin{aligned}
\mathbf{E}[T_1^{(B)} \vee T_1^{(O)}] &= \frac{1}{13} + \left(\mathbf{P}(X_1 = (B)) \cdot \mathbf{E}\left[T_1^{(O)}\right] + \mathbf{P}(X_1 = (O)) \cdot \mathbf{E}\left[T_1^{(B)}\right]\right) \\
&= \frac{1}{13} + \left(\frac{12}{13} \cdot 1 + \frac{1}{13} \cdot \frac{1}{12}\right) \\
&= 1 + \frac{1}{12 \cdot 13}
\end{aligned}$$

Grading Comments for Q1.

a. Of the three parts, the students did best on this part. Some students included Oprah in their superpositioned process, which is incorrect. Some students also found the probability that there is exactly one Trump arrival in the first five tweets, which is incorrect and lost a lot of points.

b. Many students struggled on this part, claiming that if we are given that Trump tweets first we can thin this down to the Trump process. This is not correct, and received no credit. The reason is because labels of tweets are independent of arrivals from the superposed process $N_t^{\text{all}} = N_t^T + N_t^J + N_t^O$. So telling you the label of a tweet should not affect the arrival times of tweets from N_t^{all}.

c. Many students claimed that the arrival time for both tweets was just the sum of the expected times for each tweet. This is incorrect, and was covered in both homework and precept. Others used the incorrect formula for the expectation of a maximum of exponentials. Some said that since it takes Oprah longer to tweet, we just have to wait for her to tweet, which is wrong.

Q. 2 (Smoking is bad for you)**.** Let X be the non-smoker's lifespan (with hazard rate function h) and let Y be the smoker's lifespan (with hazard rate function $2h$). They are independent, and their distributions are given by

$$\mathbf{P}\{X > t\} = \exp\left(-\int_0^t h(u)du\right)$$
$$\mathbf{P}\{Y > t\} = \exp\left(-\int_0^t 2h(u)du\right)$$

where h satisfies $\int_0^\infty h(u)du = \infty$.

a. From above we have that

$$\mathbf{P}\{X \le t\} = 1 - \exp\left(-\int_0^t h(u)du\right).$$

To get the density of the lifetime, we differentiate the CDF with respect to t

$$\mathbf{P}\{X \in dt\} = \frac{d}{dt}\mathbf{P}\{X \le t\} = \boxed{h(t)\exp\left(-\int_0^t h(u)du\right)dt}.$$

We used the fundamental theorem of calculus in the above calculation to get

$$\frac{d}{dt}\int_0^t h(u)du = h(t).$$

b. Now we can compute the probability by conditioning:

$$\begin{aligned}
\mathbf{P}\{Y > X\} &= \int_0^\infty \mathbf{P}\{Y > t | X = t\}\mathbf{P}\{X \in dt\} \\
&= \int_0^\infty \mathbf{P}\{Y > t\}\mathbf{P}\{X \in dt\} \text{ since } X \text{ and } Y \text{ are independent} \\
&= \int_0^\infty \exp\left(-\int_0^t 2h(u)du\right) h(t) \exp\left(-\int_0^t h(u)du\right) dt \\
&= \int_0^\infty h(t) \exp\left(-\int_0^t 3h(u)du\right) dt \\
&= -\frac{1}{3} \exp\left(-\int_0^t 3h(u)du\right)\Bigg|_0^\infty = \boxed{\frac{1}{3}}.
\end{aligned}$$

Grading Comments for Q2. This question was surprisingly not well answered.

a.
- The most common mistake here was solving the problem assuming that the hazard rate function is a constant—that is using h instead of $h(t)$ in the integrals. The question explicitly tells you that you may not make this assumption. Working with this assumption means that you're solving a significantly easier problem.
- Another mistake was not knowing how to use the fundamental theorem of calculus to differentiate $\int_0^t h(u)du$.
- A few students claimed that $\int_0^\infty h(u)du = 1$ instead of $\int_0^\infty h(u)du = \infty$.
- Some students wrote $\int_0^t h(t)dt$. You cannot have t as your variable of integration and also appear in the bound of the integral.

b.
- The most common pitfall here was students using thinning and superposition argument and concluding that

$$\mathbf{P}\{Y > X\} = \frac{h(t)}{2h(t) + h(t)} = \frac{1}{3}.$$

  We did not cover a thinning or superposition for nonhomogeneous Poisson processes in this class, and it is not entirely obvious or trivial that it must be true (though something like this should be true if the ratio between the rates is time-independent, as in this example, but you would have to prove it!) The thinning theorem we learned in class does not apply here when the rate function h(t) is nonhomogeneous.
- A few students knew how to do the computation all the way up to

$$\int_0^\infty h(t) \exp\left(-\int_0^t 3h(u)du\right) dt,$$

  but did not know how to complete this integral. Please ask the AIs or review your calculus notes on how to do such integrals.

- Similar to part (a), some students worked under the assumption that $h(t)$ was a constant. The spirit of this problem was to use conditioning to solve a non-homogeneous rate problem. Assuming a constant hazard rate function defeats the purpose of the question—you are solving a different, easier problem.

Q. 3 (An asymmetric game)**.** a. At each step the game as probability $\frac{1}{2}$ of ending. Let T be the step at which the game ends, then for every n:

$$\mathbb{P}(T = \infty) \leq \mathbb{P}(\text{Game lasts at least } n \text{ rounds}) = \frac{1}{2^n},$$

so $P(T = \infty) = 0$.

b. Define $\mu(n) = \mathbb{E}(T | S_0 = n)$ with boundary conditions

$$\mu(4) = \mu(0) = \mu(-1) = 0.$$

Then we have

$$\mu(3) = 1 + \frac{1}{2}\mu(1),$$
$$\mu(2) = 1 + \frac{1}{2}\mu(3),$$
$$\mu(1) = 1 + \frac{1}{2}\mu(2).$$

So $\mu(3) = \mu(2) = \mu(1) = 2$.

c. Define $f(n) = \mathbb{P}(S_T = -1 | S_0 = n)$ with boundary conditions $f(4) = f(0) = 0$ and $f(-1) = 1$.

We have

$$f(3) = \frac{1}{2}f(1),$$
$$f(2) = \frac{1}{2}f(3),$$
$$f(1) = \frac{1}{2} + \frac{1}{2}f(2).$$

So $f(2) = \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}f(2)\right)$ which implies that $f(2) = \frac{1}{7}$.

Grading Comments for Q3.

- For part a. it does not suffice to say we showed this in class. We have never covered hitting probabilities for three boundaries.
- For part a. it does not suffice to simply say it holds by Murphy's law. We must identify the recurring event first.
- For part b. the formula for random walk hitting times does not work because they are only valid for walks that take steps $+1$ or -1.
- For part c. many students calculated the probability of hitting either 0 or -1.